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Original Research Paper

The regional facet of a global innovation system: Exploring the spatiality of resource formation in the value chain for onshore wind energy

Sebastian Rohe

Working Group "Organisation & Innovation", Department for Social Sciences, Carl von Ossietzky University of Oldenburg, Ammerländer Heerstraße 114-118, 26129 Oldenburg, Germany

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ABSTRACT

Onshore wind energy can be conceptualized as a "Global Innovation System" in which resources drive technological development and diffusion: Knowledge, markets, investment and legitimacy. This paper investigates the regional embeddedness of resource formation along the industry's value chain, and explores how these processes are coupled to each other and to the national and global scale.

Empirically, the contribution focusses on a well-developed region within the GIS and highlights different spatial patterns of resource formation: Turbine manufacturing is characterized by spatially sticky knowledge and price-driven global valuation. The diffusion of the technology by downstream value chain and service-based firms reveals different processes: Knowledge for turbine management and system integration is similarly footloose and sticky. The regional level is vital for market formation (siting decisions) and technological legitimation (community acceptance). These valuation processes rest on distinct and diverse kinds of knowledge, and are interlinked, customized and facilitated by regional actor networks.

1. Introduction

To keep global warming below two degrees Celsius until the end of the century, energy systems need to transition towards renewable energy technologies (RET) and the development and diffusion of RET innovations such as wind energy must accelerate. The *Innovation System* perspective draws attention to this non-linear, multi-scalar, interactive and cumulative innovation process and the environment in which it takes place (Edquist, 1997). Within this system, knowledge is the primary resource for the development of technological innovation, while market access, financial investment and legitimacy are primary resources for the technology's valuation and diffusion (Binz et al., 2016; Musiolik et al., 2018).

The *Global Innovation Systems* (GIS) framework captures how system resources are created and distributed across places and scales and highlights how specific characteristics of a technology shape these spatial configurations. The GIS for onshore wind energy has been characterized as "production-anchored", meaning that the innovation mode for this technology is spatially sticky and locally embedded, while the system for valuation is rather footloose, with standardized global markets and a well-established technological legitimacy (Binz and Truffer, 2017). Following this rough initial categorization, there have been calls for empirical illustrations moving "the question of 'where' system resources form […] to the center stage" (Binz and Truffer, 2017). Furthermore, an emerging strand of literature is investigating different segments of complex value chains and the distinct patters of resource formation within

E-mail address: sebastian.rohe@uni-oldenburg.de.

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them (Stephan et al., 2017; Malhotra et al., 2019; Hipp and Binz, 2020).

This paper contributes twofold to the debate: First, by examining the *Oldenburger Land* (Germany), a significant locality within the GIS of onshore wind energy, I explore regional facets of resource formation, and to what extend processes are coupled to each other (horizontally) and to national and global activities (vertically). Second, I investigate how different actors from the value chain, especially from service-based downstream segments and from turbine operation and maintainance (O&M), contribute to endogenous resource formation and thus create elements of spatial stickiness within the GIS. These actors represent the – often sidelined – technological *diffusion*. Even though innovation studies are ultimately about the development and diffusion of novel technologies, technological *development*, especially through research and in the manufacturing of core components, is usually in focus (Weber and Truffer, 2017).

This paper shows that the diffusion of onshore wind strongly depends on the formation of regional *market access* and *technological legitimacy*: Suitable sites for new developments are limited and the backlog of siting decisions decreases turbine demand (market access). At the same time, legitimacy for specific wind projects must be created regionally, as "community acceptance" is the litmus test for a generally high, but diffuse (inter)national "socio-political acceptance" (Wüstenhagen et al., 2007). As both processes affect each other and emerge from the interplay of local actors, informal networks and regional institutions, these resources cannot be easily transferred across space. While *financial investment* has clearly become globalized and flows into the region through vertical coupling, the spatial dynamics for *knowledge creation and diffusion* are complex. It is difficult to capture the spatiality of knowledge formation in downstream service and O&M through the established, manufacturing-focussed dichotomy between "science and technology driven innovation" (STI) and "doing, using, interacting" (DUI) (Jensen et al., 2007). On the one hand, many firms from the region work on operating wind farms efficiently and integrating the produced electricity into grids and energy systems. While knowledge necessary for solving these issues is characterized mostly by STI, both spatially sticky and globally footloose patterns are observable. On the other hand, firms from project development, financing and operation combine entirely different and diverse kinds of knowledge. In conclusion, developing and manufacturing core components for wind turbines can be considered production-anchored (Binz and Truffer, 2017), while technological diffusion and use is characterized by a regionally embedded system for valuation and both sticky and footloose innovation dynamics.

In the next chapter, the theoretical foundations of this paper are introduced. In chapter three, the research question and scope are elaborated. After a brief methodological section in chapter four, I present insights from a case study on the well-developed wind sector in the Oldenburg region. For each resource type, I discuss how they are created regionally, and how they are coupled to each other and to national and global activities. In the last chapter, I reflect on the different facets of spatial embeddedness within the GIS and the value chain, and discuss theoretical and policy implications.

2. Onshore wind energy as a Global Innovation System

This chapter briefly introduces key debates in which GIS is rooted, before elaborating on the framework and summarizing the research around the innovation system 'wind energy'.

2.1. Theoretical background: transitions, innovation systems, and regions

The development and diffusion of RET can be conceptualized as a sustainability transition, a "fundamental transformation processes through which established socio-technical systems shift to more sustainable modes of production and consumption" (Markard et al., 2012). The notion of "socio-technical systems" goes beyond the mere technological dimension and includes corresponding, complementary or conflicting changes in institutional structures and user practices, shedding light on the interplay of actors, their networks and institutions and the products, services, business models and organizations that change along with technology (Coenen and Díaz López, 2010).

Technological Innovation Systems (TIS) is a widely used framework for analyzing transitions and the functional dynamics within an innovation system for (novel) technology. These "functions" influence system performance, and the success of technological development and diffusion (Markard and Truffer, 2008; Panetti et al., 2018). All structural components (actors, networks, institutions) actively contributing to the development of a particular technology belong to a TIS (Bergek et al., 2008).

However, TIS studies have been criticized for "interpret[ing] nation states as the natural size of TIS" (Mattes et al., 2015) and "methodological nationalism" (Coenen, 2015). In response, a spatial turn (Coenen et al., 2012) led to insightful studies comparing national TIS (Wieczorek et al., 2015), analysing cross-country influences (Bento and Fontes, 2015; Quitzow, 2015) and introducing the concept of scale-transcending TIS (Dewald and Fromhold-Eisebith, 2015).

Only few contributions investigate TIS from a regional perspective (Martin and Coenen, 2015) and take into account the specific socio-economic, institutional and geographic features of the regional context into which the TIS is embedded (Bergek et al., 2015). The term *region* is used incoherently: Sometimes 'regional' and 'local' are used interchangeably (Mattes et al., 2015); sometimes 'region' is the opposite of 'city' (Späth and Rohracher, 2012); sometimes 'regional' is an intermediary governance level between national and municipal (Dewald and Fromhold-Eisebith, 2015). In this paper, the 'region' is "a territorial unit nested between local and national level" (Faller, 2014), not primarily determined by administrative boundaries, but rather by what actors perceive as 'the region' and the scope of their regional social networks (Hansen and Coenen, 2015).

2.2. Introduction to Global Innovation Systems (GIS)

Instead of focussing on a specific scale, the *Global Innovation Systems* framework emphasizes the role of "multi-scalar networks and systemic differences between the innovation processes in various industries" (Binz and Truffer, 2017). It evolved from TIS, but combines elements from economic geography and other innovation system perspectives. Within the GIS, actor networks and institutions create system resources to carry forward the development and diffusion of a technology (or industry; both terms are used synonymously in GIS). The functions distinguished in TIS studies (Hekkert et al., 2007; Bergek et al., 2008) are condensed into four analytical system formation processes: 1. *Knowledge creation*, involving activities that create new knowledge about a technology and lead to its diffusion among actors; 2. *Market formation*, involving activities that create and grow (niche) markets or market segments; 3. *Investment mobilization*, involving activities that aim to mobilize financial capital; 4. *Technology legitimation*, involving activities that embed the technology in institutional structures or adapt institutional environments (Binz et al., 2016). These processes ultimately lead to four system resources: Knowledge, markets, investments and legitimacy.

System resources are "tangible and intangible assets of strategic value to which actors [in the innovation system] have [non-excludable] access" (Musiolik et al., 2018). They are relational and therefore collectively created (Bathelt and Glückler, 2005). While some studies describe this process as emergent and shaped by "distributed agency" (Garud and Karnøe, 2003), others highlight strategic efforts to create system resources through cooperation in formal (and informal) networks (Chetty and Agndal, 2008; Musiolik and Markard, 2011). The relational view of system resources emphazises variations in resource creation and use between territorial, social and institutional contexts, and how "positive returns from one resource type can be transferred to generate positive returns from another" (Bathelt and Glückler, 2005). In GIS, the ability of actor networks to access and combine system resources from other contexts is called "structural coupling" (Binz and Truffer, 2017).

The GIS framework shows how two specific characteristics of a technology influence the spatiality of system resources: First, there is the technology's "innovation mode", shaping knowledge formation either by STI or DUI (Jensen et al., 2007): STI relies primarily on "know-what" and "know-why", codified information usually originating from scientific research in labs and universities. DUI rather depends on tacit knowledge and transmits "know-how" and "know-who" through solution-oriented, face-to-face interaction between producers (such as engineers) and the product's users. Empirically, Jensen et al. (2007) find these two ideal-type innovation modes often occurring in combination. In terms of spatiality, technologies relying primarily on STI are associated with a more globally footloose channelling of knowledge through multi-scalar and scale-transcending networks, while the DUI mode relies primarily on locally sticky knowledge rooted in specific institutional contexts of certain regions (Binz and Truffer, 2017). Noticeably, creation and diffusion patterns of different kinds of knowledge are not only related to geographical distance and vice versa, but also to cognitive, organizational, institutional and social proximities (Boschma, 2005; Mattes, 2012).

Second, there is the "system of valuation" through which a technology becomes a valued product for its users. It draws primarily on markets, investments and legitimacy and has two valuation modes that depend on the technology's complexity and shape how resources are distributed spatially. On the one hand, there is *standardized valuation* for relatively homogenous and simple products, like fast-moving consumer goods or PV-modules. For these, a global market driven by price signals exists, and user preferences are relatively uniform and coherent across institutional contexts. On the other hand, there is *customized valuation* for complex products and product systems, like consulting services or biogas plants, that are tailored to customer demands or institutional contexts. Place-specific knowledge and complex negotiations with different stakeholders are needed to tap into new market segments (Huenteler et al., 2016b; Binz et al., 2017; Binz and Truffer, 2017).

Combining these dimensions and their respective ideal modes for innovation or valuation results in a generic typology of four GIS configurations. Each GIS type has distinct spatial resource formation patterns, which infer different policy and governance

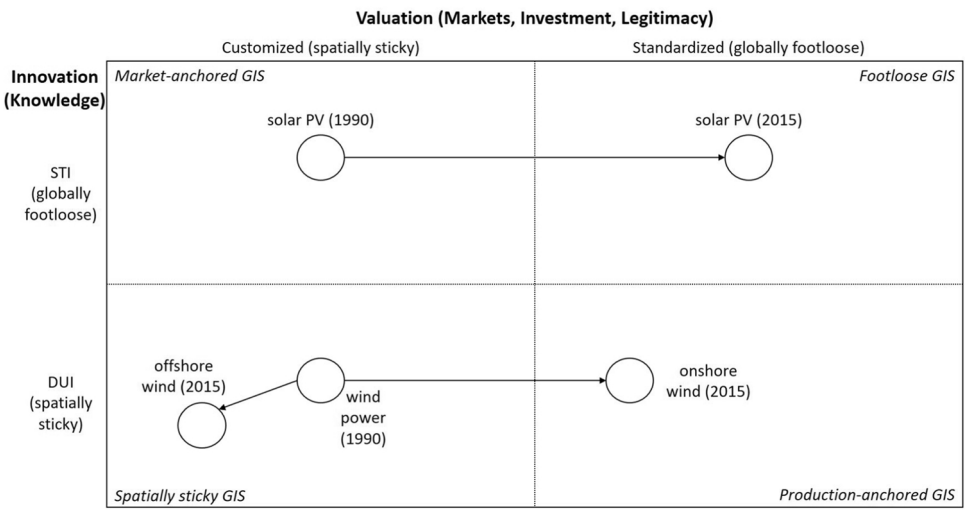

Fig. 1. Spatial configuration of different GIS types (Figure based on Binz and Truffer, 2017).

consequences (Binz and Truffer, 2017). Within this typology (cf. Fig. 1), technologies can be placed in relation to each other and their development over time. For instance, solar PV has always been characterized by an STI innovation mode (Binz et al., 2017). But it changed from a customized product, depending on regional and national formation of niches and favorable policy conditions, towards a standardized mass product with an intense global price competition (Dewald and Fromhold-Eisebith, 2015; Zhang and Gallagher, 2016).

Recently, TIS and GIS scholars moved from these broad technology-level analyses and zoomed into value chain segments (van Welie et al., 2019; Hipp and Binz, 2020). These studies often use patent data to assess knowledge development and diffusion in the manufacturing of RET, therefore largely ignoring valuation and downstream value chain segments (Stephan et al., 2017; Hoppmann, 2018). Service firms (from IT, financing or project management) and the spatial patterns of their specific innovation activities (for instance process or business innovations) are thus under-reported or not accounted for (Malhotra et al., 2019), even though they are a crucial element in the diffusion of complex, project-based technologies like onshore wind energy.

2.3. The spatiality of the innovation system for onshore wind energy

Worldwide, wind energy is the second largest RET regarding installed capacity and the second fastest growing RET regarding net additions (IRENA, 2019). The development of modern turbines began in the 1970s in Denmark and later in places such as Germany, the US and China (Karnøe and Garud, 2012; McDowall et al., 2013). The technology matured, resulting in larger plants, decreasing prices (Huenteler et al., 2016a) and a certain decoupling of onshore and offshore applications (Wüstemeyer et al., 2015). The industry is growing globally and provides around 1.2 million jobs (IRENA, 2019). The largest turbine manufacturers in 2017 were Vestas (Denmark), SiemensGamesa (Spain/Germany), Goldwind (China), GE (USA) and Enercon (Germany) (FTI Intelligence, 2018).

This overview shows how the innovation system outgrew its niche-status – at least in places like Denmark or Germany, where it originally emerged, industrial clusters have been established, and wind energy contributes 46.9 %, respectively 17.4 % to the electricity production (Agora Energiewende, Sandbag, 2019). In response to the (spatially uneven) diffusion of RET, scholars described different “phases” in the development of an innovation system: From formation and growth, to a phase of maturing, to, eventually, decline (Markard, 2018). In North-West Europe, wind energy left the formative phase (Wilson, 2012; McDowall et al., 2013; Bento and Wilson, 2016). Although GIS does not explicitly incorporate a phase model, it recognizes that technological development and diffusion and its corresponding spatial patterns change dynamically (Binz and Truffer, 2017).

Wind turbines are a “product system” with a “complex product architecture” (Huenteler et al., 2016b). The GIS used to be “spatially sticky” in the 1990s (Binz and Truffer, 2017), when DUI was the predominant innovation mode, with co-located producers and users jointly applying their experience to drive forward developments in turbine technology (Garud and Karnøe, 2003; Karnøe and Garud, 2012). Its valuation mode was customized and locally embedded, with heterogenous, volatile and fragmented markets (McDowall et al., 2013) and a “strong geographic variation in terms of specialized user needs, regulation, and levels of technology legitimacy” (Binz and Truffer, 2017).

The wind energy’s innovation mode remains – in the assessment of the turbine manufacturing at the value chain’s core by Binz and Truffer (2017) – characterized by DUI, as knowledge is still mostly generated where the technology first emerged (Lewis, 2011; Huenteler et al., 2016b). This is especially true for offshore, a relatively new application of the technology. In this sector, there is much need for experimenting and interacting between producers and users and the institutional embedding plays a key role for innovation and valuation (Wieczorek et al., 2015; Wüstemeyer et al., 2015). In the onshore case, Binz and Truffer (2017) briefly state that “turbines are now a standardized product with price driven global market competition”, implying a globally footloose valuation and characterizing the GIS as production-anchored.

3. Research question and scope of investigation

Binz and Truffer (2017) call for a thorough investigation of resource formation processes along the value chain and in different locations, exploring “what type of system resources get created where and how”. In this paper, I focus on the regional embeddedness of resource formation for onshore wind, especially in downstream value chain segments and O&M and ask:

> How do value-chain actors in the onshore wind energy GIS create system resources within a region, and to what extend are these processes coupled to each other and to national and global activities?

The following sections lay out how regional resource formation and structural couplings are operationalized, and which value chain actors and activities are primarily investigated.

3.1. Operationalizing regional resource formation and structural couplings

Some regionally available system resources might be created at another scale and are then accessible ubiquitously. For instance, the national EEG (Renewable Energy Act) regulates market access for RET across Germany. But resources might also be mobilized within specific regions, through the interplay of endogenous actor networks and institutions. One example could be a local effort on the synchronization of electricity production from wind turbines with an adjacent refrigerated warehouse. This knowledge might create local niche markets (horizontal coupling of resource formation) and trigger national regulation (vertical coupling). Thus, even if resources are primarily created regionally, through DUI innovation or customized valuation, they are not always contained within the region and might spill to other places and scales.

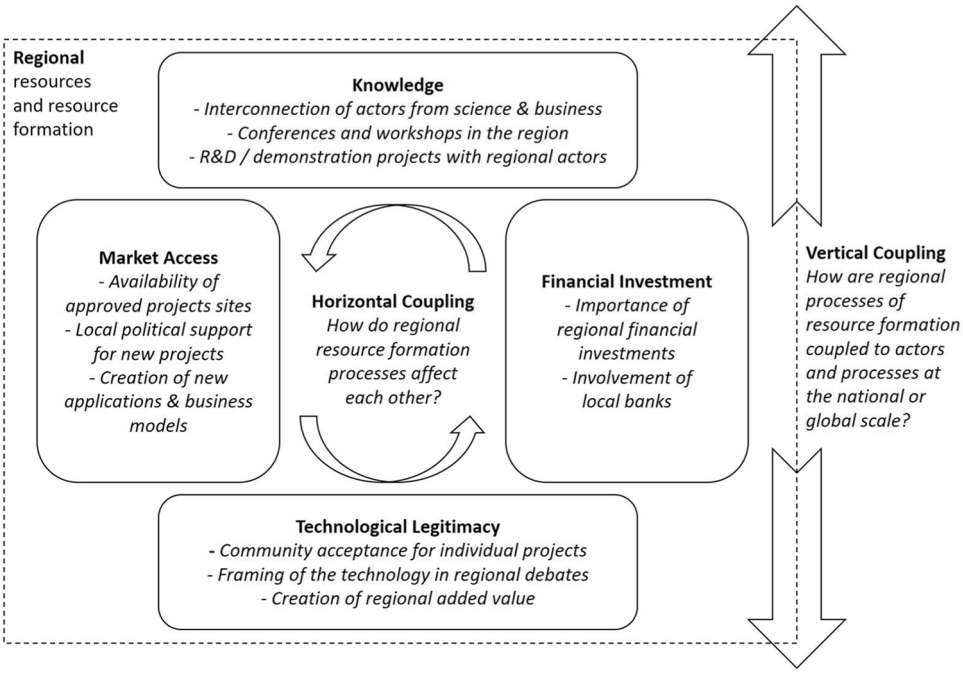

Fig. 2. Operationalizing regional resource formation and structural couplings (Own figure).

Fig. 2 summarizes the scheme for the empirical analysis. The operationalization is inspired by a range of TIS and GIS studies that used various indicators for mapping system functions and resources (Hekkert et al., 2007; Gosens and Lu, 2013; Bento and Fontes, 2015; Binz et al., 2017). As the focus of this investigation lies on regional activities, resources are mapped not primarily through national events, and quantitative measures, but rather through qualitative accounts from stakeholders and regional indicators and activities (Martin and Coenen, 2015; Wieczorek et al., 2015; Planko et al., 2017).

3.2. The value chain in onshore wind energy

Fig. 3 depicts the value chain in onshore wind energy. As discussed in Section 2.2, the value chain position in which organizations are active influences the innovation process and the spatial configuration of resource formation.

The upstream segment of the value chain provides raw material, components and equipment for turbine production, such as machinery (Liu et al., 2018), but also inputs from basic research & development (R&D) (Zhang and Gallagher, 2016). At the value chain's core, central components like blades, gearboxes, generators and towers are manufactured (Wüstemeyer et al., 2015) and

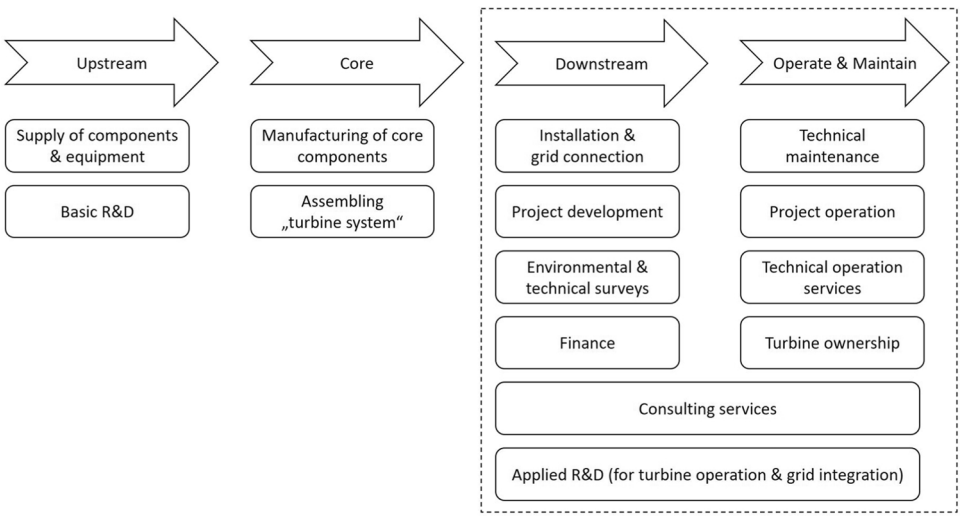

Fig. 3. Value chain in onshore wind energy (Own figure).

assembled into a "turbine system" (Huenteler et al., 2016a). Resource formation in these two segments is not at the center of this paper for two reasons: First, these processes have been discussed in various contributions already (cf. Sections 2.2 and 2.3). Second, the investigated *Oldenburger Land* mostly hosts actors from downstream and O&M segments (cf. Section 4 for a discussion of the case study selection).

The latter two value chain segments are mostly engaged with technological diffusion. Downstream, turbines are installed and connected to the grid. This requires project developers to coordinate different tasks, such as securing financial capital, collecting technical and environmental surveys in order to assess the profitability of a wind park at a specific site, and obtain necessary permits from local authorities. Once a wind farm is operating, turbines need regular technical maintenance. Project operators must manage day-to-day business, sell the produced electricity and run turbines according to provisions from regulators and grid operators. Service companies might facilitate this, for instance by providing software solutions. Consulting (for instance on legal requirements or business models) and applied R&D projects on turbine operation and grid integration are spanning both of the latter value chain segments and support technological diffusion (Malhotra et al., 2019).

4. Methodology

The research question of this contribution requires to zoom into a particular region – in this case, the *Oldenburger Land*. The study on this region was conducted in the context of the project 'Regional Energy Transitions' (REENEA). Within the project, case studies on the wind energy sector in six regions were carried out. They were selected to represent a broad and representative spectrum in terms of actor bases, histories, and social dynamics in regional wind energy deployment. Out of these regions, I chose the *Oldenburger Land* as the empirical case for this paper, because it hosts a particularly dense and regionally embedded network of various firms and organizations with diverse size and value chain position. Therefore, Oldenburg is well suited to investigate regional resource formation – at least from the perspective of a well-developed and relevant locality within the GIS.

The case study (Yin, 2009) primarily draws on 28 semi-structured face-to-face interviews with diverse actors from the Oldenburg region: Research institutes, firms for project funding, development and operation, service providers for downstream and O&M activities, but also actors from utilities, regional politics, administration and clusters. The interviews cover activities of the organization in the wind sector and the region, their cooperation with regional and extra-regional actors, the external and endogenous forces perceived to drive regional development of wind energy, and the relevance of their organization's location for their work. Additionally, I draw on six extra-regional interviews with stakeholders from other regions, the *Länder* and the national level (cf. Appendix). These additional interviews are used to triangulate the perceptions of the regional interviewees and to account for vertical coupling and multi-scalar networks reaching beyond regional borders (Hansen and Coenen, 2015). Interviews were conducted between 2018 and 2019 and took around 60 min. They were recorded, transcribed, and coded with the use of MAXQDA.

Furthermore, thematically relevant regional workshops and conferences were observed and documents, newspaper articles and webpages were studied. Interviewees and other regional stakeholders participated in a workshop, in which initial findings were presented and critically assessed by the practitioners. All these inputs informed the comprehensive case study that serves as the foundation for the following empirical discussion.

5. Resource formation in the *Oldenburger Land*

The *Oldenburger Land* had for centuries been autonomous and later constituent state within Germany. Today, the region in Lower-Saxony consists of six rural and three urban districts, the largest being the city of Oldenburg with 170,000 inhabitants. Overall, 1.1 million people live in the relatively sparsely populated area, which has a flat topography and strong winds due to its proximity to the North Sea.

The university of Oldenburg was founded in 1973 and its physics department focussed on RET research early on. In 1980, a specialized RET study program was introduced and a sophisticated energy lab for electricity system research was built. From this university nucleus emerged many of the medium-sized service firms (active in project certification, surveying, wind forecasting, consulting, etc.), research institutes and project developers that are still active in the region. In 1984, the turbine manufacturer Enercon was founded close by, and a few years after the local EWE became one of the first German utilities actively involved in the rollout of wind energy. The installation of turbines began in the 1990s and accelerated in the early 2000s and again after 2014 (cf. Fig. 4). As of 2018, around 3 % of Germany's total onshore capacity was installed in the region (while it only covers 1.5 % of the country's land area and 1.3 % of its inhabitants).[1]

5.1. Knowledge

Most interviewees suggest that the core components of the wind turbine technology matured: Potential for cost reduction is marginal and a dominant design has been established (Huenteler et al., 2016a). Knowledge creation processes still taking place in this value chain segment are characterized by DUI, as they are rather tacit and locally concentrated to the manufacturers' headquarters where engineers drive innovations (OL09, ER04, ER06). Therefore, almost no direct knowledge diffusion and R&D cooperation exists

[1] Numbers are based on a comprehensive database (The Wind Power, 2019), according to which around 51.5 GW of onshore wind were installed in 2018 in Germany; 1.5 GW within the *Oldenburger Land*.

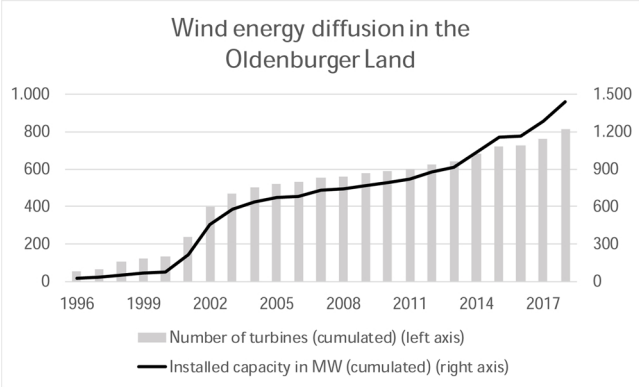

Fig. 4. Wind energy diffusion in the Oldenburger Land (Own figure).

between research institutes, service companies and project developers in Oldenburg, and the adjacent turbine manufacturer (OL02, OL08, OL24). This supports recent findings of a "reduced importance of direct interaction with the end-use sector [by manufacturers]" (Malhotra et al., 2019).

Within the region, however, there are strong **interconnections of actors from science and businesses**. Knowledge diffuses through training programs along the value chain. For instance, a course offered by a research institute and a cluster organization conveyed the skills to develop and operate wind farms to a participant from a local bank (OL26). Undertaking these tasks requires diverse types of knowledge, including a basic understanding of business and technology (know-what), experience in project management (know-how), and getting-to-know local people and customs (know-who) (OL04, OL21).

Knowledge also diffuses through dense informal networks, as many regional firms share a university-background, a history of cooperation and therefore institutional and social proximity:

It is easy. I know most of them. […] This is ideal in Oldenburg: There are almost no partners, that we would not find here (OL06).

A RET cluster was initiated by a former major in the mid-2000s and further facilitates "local buzz" (Bathelt et al., 2004). It enables actors new to the region or from adjacent economic sectors to participate in the creation and diffusion of knowledge (OL05, OL15). The cluster frequently organizes **conferences** to discuss technological solutions and business models, and facilitates **joint R&D and demonstration projects** (OL10, OL16). Actors within these projects perceive rigid national regulatory conditions as the biggest obstacle for more effective regional experimentation.

Generally, local actors from downstream and O&M value chain segments view the integration of wind power into energy systems as the greatest future challenge (OL02, OL14, OL17), and they are working together on this in several projects:

If we really want to succeed with the Energiewende, we need to test it in a larger region. […] But as a precondition, we need to have a wind energy production of more than 100 % – we have that here (OL02).

Innovating actors in Oldenburg do not solely obtain knowledge for solving this challenge from within the region, but also through **vertical coupling**: They are embedded into international research communities (OL02), professional networks (OL03) and demonstration projects (OL06). However, actors from research or services commonly refer to their home region as an "*experimentation field*" and "*laboratory*" (OL18). As these organizations create solutions locally, they codify the knowledge and transfer it to other locations and applications, for instance through software, business models (OL20), research papers or an online study program for international master students (OL21):

We can ship most of our services [electronically across the world]. Of course we have to go there from time to time, but we don't have to have an office somewhere [because] we don't need it (OL18).

These activities reflect a growing trend of innovation in turbine operation (Malhotra et al., 2019), that require competences in natural and computer sciences typically associated with STI. This innovation mode characterizes the work of the service providers, survey companies and research institutes from Oldenburg. While STI-driven innovation is potentially footloose and globally transferable, it also shows a sticky facet in this case, as actors profit from the dense and complementary network and favorable framework conditions in their region. This facilitates both endogenous knowledge creation and knowledge flows in and out of "global pipelines" (Bathelt et al., 2004).

5.2. Markets

The economic competitiveness of their turbines determines the success of manufacturers at the core of the value chain that need to sell their products downstream. Which turbines are bought in the end by developers is "*ultimately a question of price*" (OL27):

For larger projects, we invite manufacturers to an international tender. Then we compare: At first the economic efficiency and prices of the turbines, but also of course the technological feasibility, the availability and the track record (ER05).

While this indicates a standardized and footloose validation mode, the process through which developers then construct their markets by securing permits and sites is place- and project-specific (Gailing and Röhring, 2015) and thus highly customized.

Generally, regulations or an interplay of entrepreneurial push and customer demand drive market creation (Planko et al., 2017). National provisions like feed-in-tariffs or tenders are often used as indicators for market formation in the RET sector (Quitzow, 2015; Binz et al., 2016, 2017). One such regulation is the German EEG, according to which developers need to successfully bid in a national tender to receive a subsidy. Beforehand, developers must have secured sites, obtained surveys and a pre-approval from local authorities (Lieblang, 2018). Between 2014 and 2018, the capacity of pre-approved onshore wind projects dropped around two thirds (FA Wind, 2018). Accordingly, the participation in national tenders has been decreasing and fell below the overall tender volume (BNetzA, 2019). The **lack of pre-approved project sites** thus became a major bottleneck to the rollout of onshore wind energy in Germany (ER03, ER05). This barrier must be overcome at a regional level, as the process of receiving pre-approvals is driven by the interplay of entrepreneurs (project developers), municipal governments (needing to grant pre-approval) and regional stakeholders like land owners and survey firms (OL08, OL09, OL17). In the *Oldenburger Land*, some wind projects are not moving forward, as they are stuck in prolonged administrative processes and lawsuits filed by local initiatives and associations (OL27, OL28).

To overcome these slumps in regional siting decisions, all interviewed developers stressed their need to be embedded into regional networks and have good relations to both local planning authorities (OL27) that have "*leeway as to how quickly and thoroughly things go*" (OL23), and land owners. Therefore, downstream segments of the wind sector have been commonly referred to as a "*handshake-business"* where "*contacts in the region make all the difference"* (OL23):

> *[In the late evening], we sat down in the kitchen of one of the land-owners [and] drank a 'Bullenschluck' schnaps, because I had made a final suggestion. The land owners said: 'Yes, we do it like this.' [So] we cut the knot and got the signature. That wasn't easy, because they said: 'We have the area. It won't work without our site.' But we told them: 'If politics say it won't work, then the site won't help you.' [So we all] had to reach an understanding (OL26).*

Due to the importance of local ties for securing access to sites and therefore markets, many developers operate in a limited geographical range, "*where you know the way around, speak the local dialect and know the people"* (OL09):

> *The geographical distance is one factor, but so are […] the different ideas and mind-sets [in other regions]. When it comes to acquisition it is important that you can convey the promise and that certainly doesn't work so well in Baden-Württemberg and Bavaria (OL04).*

A trustful relationship between public authorities and survey firms also facilitates planning processes:

> *Much more important is the relationship with the nature conservation authority. […] The project planner pays us, but the [local] authority depends on us writing a reasonable expert opinion […] When they know'Well, [this firm does] sensible work, I can simply read the summary and know what's going on', then that makes their work easier (OL12).*

However, overall **political and administrative support for new projects** is dwindling, as regional decision makers increasingly dread negotiating and designating new areas for wind development. This is caused mainly by growing civil protest and waves of lawsuits usually accompanying the planning procedures, but also by a lack of information and personal capacity in regional governing bodies (OL28). Therefore, some local administrators call for more rigid, top-down planning and siting provisions, as other federal states have imposed (Goetzke and Rave, 2016). In Lower-Saxony, planning competences remain with municipal and district administrations. The state government argues that this policy enables regional decision makers to solve conflicts where they occur (ER03).

The high regionally installed capacity of wind energy propels the increasing attention towards system-related issues (chapter 5.1) in Oldenburg and actors perceive it important to create new (niche) markets (OL17, OL25). As they seek to use the abundance of electricity generated from wind turbines, **new business models** for storage, sector coupling or marketing of regional electricity develop. Using green energy to produce hydrogen has been identified as a core strategy for regional growth (OL10, ER02). In order to harvest these business opportunities, the importance of local innovators, electricity producers and consumers to cooperate increases (OL15, OL23). However, national framework conditions hamper the profitability of business models:

> *All of us [half a dozen wind farm operators and the regional utility] are located so close together, we said'Well, we have the possibility to place and operate a large storage facility'. But some are still a little reluctant and said'It doesn't really pay off yet'(OL26).*

> *That's an interdependence. The [different scales] depend on each other. You can't [implement a business model] regionally that you haven't fed back nationally or internationally, and you have to take on board what comes internationally and nationally in the form of regulations (OL24).*

These quotes highlight the regulatory co-evolution and **vertical coupling** of market formation between the regional and the national level: In order for increased national ambitions or tendered capacities to be effective, they must match a regional willingness or preference to implement sector coupling solutions or disclose new turbine sites, and vice versa.

Vertical coupling is also visible in project planning. During some stages of the planning cycle or for specific technical surveys (regarding noise or wind conditions), regional embeddedness is less of an issue and tasks may well be solved in distanced headquarters of larger project developers or by specialized survey firms (OL03, OL09, ER05). But as highlighted above, regional embeddedness is crucial in other tasks and stages of the planning cycle. Thus, if wind developers from one region want to advance projects in other places and (inter)nationally, they often establish outlet offices as a "*mouthpiece*" because "*contacts in a region […] actually make a difference*" (OL23, OL05, OL27).

5.3. Investments

When the wind energy rollout in Oldenburg started between the late 1980s and 1995, mobilizing **regional financial investment** from a diverse range of actors was crucial. Farmers (OL24), idealistic project developers (OL09, OL04) and local banks (OL15) were among the first investors, then followed by the regional utility, that also provided a loan to the adjacent turbine manufacturer (OL14):

We built our first wind farm [in 1992], and it was very difficult. Nobody wanted to invest in wind energy. […] It was supposed to cost 20 million. An announced subsidy from Lower Saxony never came […]. [Only few 'Pioneers' and 'early bird investors' were interested and said]:'Oh yes, it might be good for the environment. You can also save taxes, but it's damn risky. […] I participate in such a wind farm, in a limited partnership and also give a little money […]'. That's how these wind farms were financed. […] Back then, it was all brave (OL09).

Still today, much of the installed wind capacity in Germany is owned by private investors and farmers (41 %), project development firms (23 %) and local utilities (11 %) (trend:research, 2017). In Oldenburg, regional companies and citizens invest disproportionally more into wind farms in their region than compared to places like Eastern Germany (OL05, OL26) (The Wind Power, 2019).

However, financing wind farms is not "*brave*" anymore. Instead, it became a profitable investment. Accordingly, it is more **difficult for local banks to become involved** in wind park financing, as they lack the capital and expertise to compete with larger firms that have specialized in financing onshore wind projects in Germany and Europe (OL05, OL27). Furthermore, international funds increasingly mobilize investments and "*buy up all the projects that appear somewhere*" (OL25), thus diminishing the ability and need for regional investors to invest. This increases the **vertical coupling** of wind farm development and operation to global financial markets (ER05):

If you have wind farms and keep them for 25 or 30 years, you have a very significant capital commitment […]. There are […] life insurance companies or institutional investors who are currently looking more for investment opportunities and have relatively low expectations regarding the returns (OL23).

Large utilities also started buying up readily developed wind projects (OL17, ER05) and downstream service firms (OL06, OL08), which further indicates footloose investment dynamics within the GIS.

5.4. Legitimacy

Interviewees unanimously report a declining acceptance of wind turbines within the region. This could potentially have detrimental effects on the diffusion of the technology (Markard et al., 2016). While the general acceptance of onshore wind is relatively high, resistance against new developments is becoming more frequent and organized (Agora Energiewende, 2018). This **community acceptance** is inherently a local resource, depending on trust among involved stakeholders, as well as procedural and distributional justice (Wüstenhagen et al., 2007).

In terms of procedural justice, spatial planning and approval processes are not sufficient anymore to create legitimacy, as all involved parties often do not find a common ground for a discussion:

In the past, facts and arguments were used to make progress [in planning and siting]. That is no longer the case. [Most of the citizens] do not really want to know what is happening. If you can't get any further with facts, then such events are no longer fun. We can only say:'You can find it stupid or not, I still want to push [the project] through' (OL04).

While project developers regard the interaction with residents as a dreadful compulsory exercise, wind farm opponents often perceive approval procedures as unfair and rigged (OL13). The clearly visible growth in turbine size outweighs clearance certificates from survey experts and other technological advancements, and thus delegitimizes the technology further (OL03, OL09). Therefore, turbine manufacturers can do very little to enhance community acceptance.

Some project developers create legitimacy by establishing long-term relationships with local communities. This requires them to make reliable commitments and remain involved in a region after the farm is put into operation – which other developers avoid for economic reasons, hence depleting the technology's legitimacy (OL13, OL23). Direct financial participation of citizens in wind energy is a way to distribute returns more fairly and increase acceptance (OL05, OL17, OL25).

A basis of trust evolved over the years. The citizens saw that there is a regional project planner: '[…] We know the people, we know the actors, we have confidence in them.' [In the second regional wind project], not even a complaint, not even a citizens' initiative. Instead, 1.400 residents wanted to invest (OL26).

In contrast to many other players in the industry, we place a lot of emphasis on regional value creation. […] Most wind farms have something to offer in that direction. We locate the wind farm companies in the municipalities. This means that trade taxes are paid locally. That is far from self-evident (OL23).

Politicians and newspapers are **framing the technology in regional debates** increasingly negative. The region is perceived to be "*built up*" (OL10, OL19) and the need for a further rollout seen as a "*burden*" (OL28). Some interviewees called for a shift in this local discourse: Rural districts should view the utilization of wind energy as an economic opportunity to supply surrounding cities with electricity (OL23, OL24). The amount of **regional value created** through wind energy varies, however, greatly between projects (Gottschalk et al., 2016). Local politicians lack legislative competence to demand community participation and often depend on the good will of developers and operators (OL22, OL28).

Ultimately, legitimacy created at the regional level might diffuse to the national level through **vertical coupling** and influence the political framework conditions for the wind energy industry:

If we have no acceptance in the region, then we will also have no acceptance in Berlin to implement the political demands [for more wind energy development] (OL17).

A similar dynamic of vertical coupling has been discussed by Dewald and Truffer (2012), who argue in the case of PV that "the EEG would barely have been possible on a national scale if legitimacy and political support had not been created in specific federal states". Accordingly, the increasing struggle to create and maintain legitimacy regionally – even, or especially, in places like Oldenburg, were the technology has matured and diffused – could transcend to the national level, for instance through frustrated regional decision makers lobbying for more restrictive rules in planning and siting (OL22).

5.5. Coupling of resources in the region

The conditions for a **horizontal coupling** of resources are beneficial in the Oldenburg region: A long-established, relatively dense actor network of firms, especially in downstream and O&M value chain segments, makes resources accessible. For instance, the research-oriented spin-offs gain (international) market access and claim legitimacy based on the knowledge mobilized within the region:

[Local R&D projects are a] reference. It is not vital for us to survive, but we have a good feeling that we have this tailwind […]. We now started in India, we are market leaders in the USA, we want to continue in Central and South America, in South Africa […]. Of course, we always try to take this positive image of the energy system transformation in Oldenburg with us. […] That opens incredibly many doors, which means that this positive background is extremely important (OL18).

Finance- and consulting firms indirectly profit from ties to local research institutes, as they get an overview of developments in the sector and future investment opportunities:

I cannot say that [our bank] acquired a lot of business [through the regional cluster]. It is an exchange:'Is there anything new? What are you working on? […]'[Ultimately] there are all small germ cells and ideas from which something larger, a viable business model, can develop, which can then be scaled. […] For this, the way in which economic performance and those who have the [necessary] intellectual input are networked plays a role. This is where [the cluster] becomes important (OL15).

In terms of the technology's diffusion, the degree to which legitimacy is mobilized in the region affects the availability of sites and therefore market formation. At the same time, if regional actors actively participate in financing, planning and utilizing wind energy, legitimacy is enhanced. Therefore, financial investment from regional stakeholders positively affects the formation of markets and legitimacy. Generally, the medium-sized, locally-rooted project developers and service firms in Oldenburg are well equipped to tap into local networks and hence couple resources for the diffusion of wind energy (OL15). However, financial investment is increasingly mobilized from outside the region and driven by European interest rates and global financial markets. This mismatch to an otherwise customized valuation might affect regional market formation and legitimation negatively (OL26).

6. Conclusion and implications

In the last section, I summarize this paper's results and their implications for the GIS framework and related theoretical debates. Furthermore, I reflect on recommendations for policy.

6.1. Summary

This contribution zoomed into the *Oldenburger Land* as an important locality within the GIS for onshore wind energy, and observed the spatial configurations of resource formation. Furthermore, the paper zoomed into different value chain segments and their specific innovation and valuation processes. For turbine manufacturing (technological development), empirical results support initial characterizations of onshore wind as production-anchored (Binz and Truffer, 2017) and an increasingly standardized technology, which market access and legitimacy depend on price signals (Lewis, 2014). Downstream and O&M value chain segments (technological diffusion), however, are clearly distinct and rely on customized valuation: Wind developers, operators and service firms create market access and legitimacy in spatially sticky processes, which are facilitated through formal and informal networks among regional stakeholders.

Regarding the innovation mode (STI vs. DUI) in technological diffusion and its spatial embeddedness, the empirically observed patterns also clearly differ from the DUI mode in manufacturing. With regards to technical surveys, operation services, consulting and applied R&D, the demand for knowledge on grid integration, data management and turbine operation increases. Knowledge required to solve these issues is primarily STI-driven (natural sciences, codifiable) and certainly has footloose qualities, as it flows into and out of the region through vertical couplings. At least in lighthouse cases like Oldenburg, however, this knowledge also has spatially sticky elements, as it is derived from and embedded into pilot projects and application-oriented networks. With regards to project development, financing and operation, innovation is even more difficult to capture empirically with the STI/global vs. DUI/local dichotomy. For these activities, the GIS frameworks' distinction between innovation and valuation becomes somewhat blurry empirically, because "know-who" and innovations in project development approaches or business models are directly connected to the

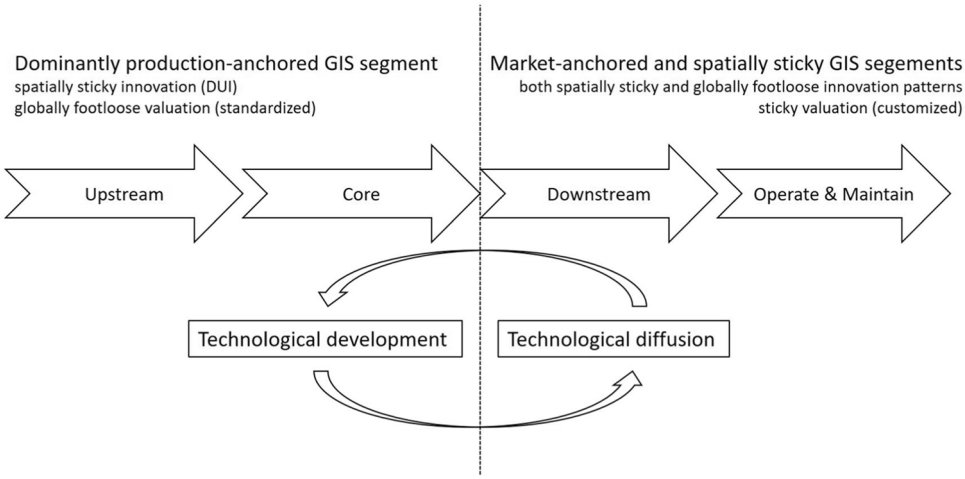

Fig. 5. Facets of a GIS – The case of onshore wind energy (Own figure).

creation of resources for valuation.

Fig. 5 summarizes the spatial configurations within the GIS for onshore wind energy. The investigation highlighted that resource formation within a GIS has multiple spatial patterns, depending on where – in which place and in which value chain segment – they take place.

6.2. Research implications

By investigating place- and value chain specific resource formation for onshore wind energy, this paper contributes to the discussion around the spatiality of technological innovation and diffusion. While the GIS framework proved suitable and helpful overall for this analysis, its weakness and the resulting research gaps are also discussed here.

Using a 'global' framework, this contribution undertook a 'regional' case study of a well-developed locality within the GIS for onshore wind. While this might sound counter-intuitive, the framework made it possible to track multi-scalar interactions and vertical couplings of regional resource formation. The framework could also be used for comparing regions that represent different development phases within the innovation system, and their distinct resource formation patterns. The GIS framework also proved applicable for value chain analyses, as it clearly distinguishes system actors from system level resources, elements that have been mixed in other scale-transcending innovation frameworks (Dewald and Fromhold-Eisebith, 2015).

This contribution revealed clearly distinct spatial patterns of resource formation between technological development and diffusion. As turbine manufacturing, and downstream and O&M value chain segments are centered around the same technology, these distinct patterns are somewhat surprising from the perspective of GIS, which ascertains that technological characteristics determine resource formation (Binz and Truffer, 2017). Differences could be explained by the particularities of the tasks, products and services that diverse actors provide. While investigated downstream firms mainly offer services, turbine manufacturers produce and assemble physical goods. The *service/goods* dichotomy might, however, not be the determining characteristic, as Binz and Truffer (2017) illustrate that all types of spatial resource formation are possible for service-based GIS.

I thus hypothezise that technologies with a *complex installation and use* (including some 'typical' spatially sticky GIS like offshore wind or biogas) involve two distinct occasions for innovation and valuation: As shown here for onshore wind, innovation and valuation not only takes place 'at the factory' and between researchers, upstream services, manufacturers and companies buying turbines (technological development). It occurs a second time during the complex installation and operation of the turbines, when other kinds of firms are involved and draw on distinct resources (technological diffusion). This is different for PV (but also other 'typical' spatially sticky GIS like watchmaking), where installation and use are very simple, and studies have, accordingly, concentrated on technological development to assess resource formation for the entire innovation system (Hoppmann, 2018; Hipp and Binz, 2020). Furthermore, decentralized technologies with a perceived or objective *unequal spatial distribution of benefits and externalities* (e.g. combatting global warming vs. local noise pollution from wind farms) might also warrant a separate innovation and valuation for their diffusion, which might be inherently more local. Future studies could test these hypotheses, and shed more light on key characteristics that determine spatial configurations in technological diffusion for different technologies, and whether and why they differ from patterns in technological development.

In order to assess the spatiality of knowledge within a GIS, the framework differentiates between two innovation modes and equates them with local (DUI) and global (STI) patterns. While these concepts proved useful to capture knowledge creation and diffusion for technological development, problems arise when it comes to empirically describing knowledge flows in downstream and service-based activities. Instead of just investigating geographical elements, future contributions could take into account other types of proximities (Boschma, 2005) and how they relate to different kinds of knowledge (Mattes, 2012) needed to diffuse complex technologies. Integrating these insights might be a promising theoretical addition to the GIS framework, and the literature on multi-

scalar innovation systems and value-chain specific resource formation.

6.3. Policy implications

Recognizing the distinct spatial configurations within a GIS can inform place-specific policy solutions. In locations hosting turbine manufacturers, policy makers should follow a production-driven support strategy by strengthening industry-supplier-university networks (Binz and Truffer, 2017). In regions with a service-oriented, downstream actor base, where issues of wind farm management and system integration are most pressing in terms of knowledge creation, policy makers might best support the industry through a lead market strategy (Binz and Truffer, 2017): Regional networks and pilot projects need leeway in the national regulatory framework to experiment with solutions on the interface between a wind farm and the energy system. This would improve the quality and scope of high-profile regional experiments, as well as their ability to create new niche markets. Regional administrators and cluster organizations should not only foster ties between regional actors, but also facilitate conditions for businesses and applied research institutes to export the potentially footloose knowledge globally.

In terms of technological diffusion, customized valuation is more important than previously accounted for (Binz and Truffer, 2017). This holds true for onshore wind, and presumably for other technologies that are complex in their installation and use and/or have a (perceived) imbalance of ubiquitous benefits and localized externalities (cf. section 6.2). This should be reflected in policy by recognizing that the regional level is the most appropriate scale at which market access and legitimacy for technological diffusion should be mobilized. Too much top-down involvement in regional planning might be counter-productive (Bergek, 2010; Fast and Mabee, 2015). Instead, the national level should provide regional decision makers with clear guidelines for the planning processes and enable them to mobilize local investments and enhance the regional value added. Such a refined governance setting would be better aligned to the place-specific valuation mode in technological diffusion for onshore wind energy.

Declaration of Competing Interest

None.

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Appendix A. Overview of Interviews

All interviews were conducted between 2018 and 2019 and lasted from 50 to 120 min.

List of interviews with stakeholders from the Oldenburg region

Code	Background/Sector of the interview partners' organization (Some interviews represent two organizations)
OL01	Regional politics & administration
OL02	Research institute (for basic and applied research)
OL03	Environmental and technical surveys
OL04	Project development and operation
OL05	Finance
OL06	Services (for downstream and O&M activities)
OL07	Boundary Organization (clusters, networks, associations)
OL08	Research institute (for basic and applied research)
OL09	Project development and operation
OL10	Boundary Organization (clusters, networks, associations)
OL11	Civil Society
OL12	Environmental and technical surveys
OL13	Civil Society
OL14	Energy Utility
OL15	Finance
OL16	Boundary Organization (clusters, networks, associations)
OL17	Services (for downstream and O&M activities)
OL18	Services (for downstream and O&M activities)
OL19	Boundary Organization (clusters, networks, associations)
OL20	Research institute (for basic and applied research)

OL21	Research institute (for basic and applied research)
OL22	Regional politics & administration
OL23	Project development and operation
OL24	Regional politics & administration
OL25	Boundary Organization (clusters, networks, associations) // Consulting services
OL26	Project development and operation // Finance
OL27	Project development and operation
OL28	Regional politics & administration

List of interviews with extra-regional stakeholders

Code	Background/Sector of the interview partners' organization
ER01	Civil Society
ER02	Boundary Organization (clusters, networks, associations)
ER03	Politics & Administration
ER04	Manufacturing of core components
ER05	Project development and operation
ER06	Manufacturing of core components

References

Agora Energiewende, 2018. Wie weiter mit dem Ausbau der Windenergie? Zwei Strategievorschläge zur Sicherung der Standortakzeptanz von Onshore Windenergie, Berlin.

Agora Energiewende, Sandbag, 2019. The European Power Sector in 2018: Up-to-Date Analysis on the Electricity Transition. Agora Energiewende. https://www.agora-energiewende.de/fileadmin2/Projekte/2018/EU-Jahresauswertung_2019/Agora-Energiewende_European-Power-Sector-2018_WEB.pdf.

Bathelt, H., Glückler, J., 2005. Resources in economic geography: from substantive concepts towards a relational perspective. Environ. Plan. A 37, 1545–1563. https://doi.org/10.1068/a37109.

Bathelt, H., Malmberg, A., Maskell, P., 2004. Clusters and knowledge: local buzz, global pipelines and the process of knowledge creation. Prog. Hum. Geogr. 28, 31–56. https://doi.org/10.1191/0309132504ph469oa.

Bento, N., Fontes, M., 2015. Spatial diffusion and the formation of a technological innovation system in the receiving country: the case of wind energy in Portugal. Environ. Innov. Soc. Transit. 15, 158–179. https://doi.org/10.1016/j.eist.2014.10.003.

Bento, N., Wilson, C., 2016. Measuring the duration of formative phases for energy technologies. Environ. Innov. Soc. Transit. 21, 95–112. https://doi.org/10.1016/j.eist.2016.04.004.

Bergek, A., 2010. Levelling the playing field? The influence of national wind power planning instruments on conflicts of interests in a Swedish county. Energy Policy 38, 2357–2369. https://doi.org/10.1016/j.enpol.2009.12.023.

Bergek, A., Jacobsson, S., Carlsson, B., Lindmark, S., Rickne, A., 2008. Analyzing the functional dynamics of technological innovation systems: a scheme of analysis. Res. Policy 37, 407–429. https://doi.org/10.1016/j.respol.2007.12.003.

Bergek, A., Hekkert, M., Jacobsson, S., Markard, J., Sandén, B., Truffer, B., 2015. Technological innovation systems in contexts: conceptualizing contextual structures and interaction dynamics. Environ. Innov. Soc. Transit. 16, 51–64. https://doi.org/10.1016/j.eist.2015.07.003.

Binz, C., Truffer, B., 2017. Global Innovation Systems: a conceptual framework for innovation dynamics in transnational contexts. Res. Policy 46, 1284–1298. https://doi.org/10.1016/j.respol.2017.05.012.

Binz, C., Truffer, B., Coenen, L., 2016. Path creation as a process of resource alignment and anchoring: industry formation for on-site water recycling in Beijing. Econ. Geogr. 92, 172–200. https://doi.org/10.1080/00130095.2015.1103177.

Binz, C., Gosens, J., Hansen, T., Hansen, U.E., 2017. Toward technology-sensitive catching-up policies: insights from renewable energy in China. World Dev. 96, 418–437. https://doi.org/10.1016/j.worlddev.2017.03.027.

BNetzA, 2019. Statistiken zum Ausschreibungsverfahren für Windenergieanlagen an Land: Statistics regarding the tenders for onshore wind energy of the Germany Federal Network Agency. Bundesnetzagentur. https://www.bundesnetzagentur.de/DE/Sachgebiete/ElektrizitaetundGas/Unternehmen_Institutionen/Ausschreibungen/Wind_Onshore/BeendeteAusschreibungen/BeendeteAusschreibungen_node.html;jsessionid=7394B3FD31C0247952A55FA8DB270C2C.

Boschma, R., 2005. Proximity and innovation: a critical assessment. Reg. Stud. 39, 61–74. https://doi.org/10.1080/0034340052000320887.

Chetty, S., Agndal, H., 2008. Role of Inter-organizational networks and interpersonal networks in an industrial district. Reg. Stud. 42, 175–187. https://doi.org/10.1080/00343400601142696.

Coenen, L., 2015. Engaging with changing spatial realities in TIS research. Environ. Innov. Soc. Transit. 16, 70–72. https://doi.org/10.1016/j.eist.2015.07.008.

Coenen, L., Díaz López, F.J., 2010. Comparing systems approaches to innovation and technological change for sustainable and competitive economies: an explorative study into conceptual commonalities, differences and complementarities. J. Clean. Prod. 18, 1149–1160. https://doi.org/10.1016/j.jclepro.2010.04.003.

Coenen, L., Benneworth, P., Truffer, B., 2012. Toward a spatial perspective on sustainability transitions: special section on sustainability transitions. Res. Policy 41, 968–979.

Dewald, U., Fromhold-Eisebith, M., 2015. Trajectories of sustainability transitions in scale-transcending innovation systems: the case of photovoltaics. Environ. Innov. Soc. Transit. 17, 110–125. https://doi.org/10.1016/j.eist.2014.12.004.

Dewald, U., Truffer, B., 2012. The local sources of market formation: explaining regional growth differentials in German photovoltaic markets. Eur. Plan. Stud. 20, 397–420. https://doi.org/10.1080/09654313.2012.651803.

Edquist, C., 1997. Systems of Innovation: Technologies, Institutions and Organizations, 1st ed. XIV, 432 S.

Faller, F., 2014. Regional strategies for renewable energies: development processes in greater Manchester. Eur. Plan. Stud. 22, 889–908. https://doi.org/10.1080/09654313.2012.741572.

Fast, S., Mabee, W., 2015. Place-making and trust-building: the influence of policy on host community responses to wind farms. Energy Policy 81, 27–37. https://doi.org/10.1016/j.enpol.2015.02.008.

FTI Intelligence, 2018. Global Wind Market Update - Demand & Supply 2017: Part One - Supply Side Analysis. https://www.fticonsulting.com/~/media/Files/us-files/intelligence/intelligence-research/global-wind-market-update-2017-supply-side-analysis.pdf. .

Gailing, L., Röhring, A., 2015. Was ist dezentral an der Energiewende? Infrastrukturen erneuerbarer Energien als Herausforderungen und Chancen für ländliche Räume. Raumforsch. Raumordn. 73, 31–43. https://doi.org/10.1007/s13147-014-0322-7.

Garud, R., Karnøe, P., 2003. Bricolage versus breakthrough: distributed and embedded agency in technology entrepreneurship. Res. Policy 32, 277–300. https://doi.org/10.1016/S0048-7333(02)00100-2.

Goetzke, F., Rave, T., 2016. Exploring heterogeneous growth of wind energy across Germany. Util. Policy 41, 193–205. https://doi.org/10.1016/j.jup.2016.02.010.

Gosens, J., Lu, Y., 2013. From lagging to leading? Technological innovation systems in emerging economies and the case of Chinese wind power. Energy Policy 60, 234–250. https://doi.org/10.1016/j.enpol.2013.05.027.

Gottschalk, M., Hoppenbrock, C., Kucharczak, L., Schäfer, S., Heike, W., Wilkens, I., 2016. Regionale Wertschöpfung in der Windindustrie am Beispiel Nordhessen. Universität Kassel. https://www.uni-kassel.de/fb07/fileadmin/datas/fb07/5-Institute/IVWL/Wetzel/Regionale_Wertsch%C3%B6pfung_in_der_Windindustrie.pdf.
Hansen, T., Coenen, L., 2015. The geography of sustainability transitions: review, synthesis and reflections on an emergent research field. Environ. Innov. Soc. Transit. 17, 92–109. https://doi.org/10.1016/j.eist.2014.11.001.
Hekkert, M.P., Suurs, R.A.A., Negro, S.O., Kuhlmann, S., Smits, R.E.H.M., 2007. Functions of innovation systems: a new approach for analysing technological change. Technol. Forecast. Soc. Change 74, 413–432. https://doi.org/10.1016/j.techfore.2006.03.002.
Hipp, A., Binz, C., 2020. Firm survival in complex value chains and global innovation systems: evidence from solar photovoltaics. Res. Policy 49, 103876. https://doi.org/10.1016/j.respol.2019.103876.
Hoppmann, J., 2018. The role of interfirm knowledge spillovers for innovation in mass-produced environmental technologies: evidence from the solar photovoltaic industry. Organ. Environ. 31, 3–24. https://doi.org/10.1177/1086026616680683.
Huenteler, J., Ossenbrink, J., Schmidt, T.S., Hoffmann, V.H., 2016a. How a product's design hierarchy shapes the evolution of technological knowledge: evidence from patent-citation networks in wind power. Res. Policy 45, 1195–1217. https://doi.org/10.1016/j.respol.2016.03.014.
Huenteler, J., Schmidt, T.S., Ossenbrink, J., Hoffmann, V.H., 2016b. Technology life-cycles in the energy sector: technological characteristics and the role of deployment for innovation. Technol. Forecast. Soc. Change 104, 102–121. https://doi.org/10.1016/j.techfore.2015.09.022.
IRENA, 2019. Global Wind Energy Statistics. IRENA. http://www.irena.org/wind.
Jensen, M.B., Johnson, B., Lorenz, E., Lundvall, B., 2007. Forms of knowledge and modes of innovation. Res. Policy 36, 680–693. https://doi.org/10.1016/j.respol.2007.01.006.
Karnøe, P., Garud, R., 2012. Path Creation: co-creation of heterogeneous resources in the emergence of the Danish wind turbine cluster. Eur. Plan. Stud. 20, 733–752. https://doi.org/10.1080/09654313.2012.667923.
Lewis, J.I., 2011. Building a national wind turbine industry: experiences from China, India and South Korea. IJTG 5, 281. https://doi.org/10.1504/IJTG.2011.039768.
Lewis, J.I., 2014. Industrial policy, politics and competition: assessing the post-crisis wind power industry. Bus. polit. 16, 511–547. https://doi.org/10.1515/bap-2014-0012.
Lieblang, L.A., 2018. Ein Jahr danach: Erste empirische Ergebnisse der Ausschreibungen für Windenergie an Land gemäß EEG 2017. Z Energiewirtsch 42, 315–328. https://doi.org/10.1007/s12398-018-0232-4.
Liu, J., Wei, Q., Dai, Q., Liang, C., 2018. Overview of wind power industry value chain using diamond model: a case study from China. Appl. Sci. 8, 1900. https://doi.org/10.3390/app8101900.
Malhotra, A., Schmidt, T.S., Huenteler, J., 2019. The role of inter-sectoral learning in knowledge development and diffusion: case studies on three clean energy technologies. Technol. Forecast. Soc. Change 146, 464–487. https://doi.org/10.1016/j.techfore.2019.04.018.
Markard, J., 2018. The life cycle of technological innovation systems. Technol. Forecast. Soc. Change. https://doi.org/10.1016/j.techfore.2018.07.045.
Markard, J., Truffer, B., 2008. Technological innovation systems and the multi-level perspective: towards an integrated framework. Res. Policy 37, 596–615. https://doi.org/10.1016/j.respol.2008.01.004.
Markard, J., Raven, R., Truffer, B., 2012. Sustainability transitions: an emerging field of research and its prospects. Res. Policy 41, 955–967. https://doi.org/10.1016/j.respol.2012.02.013.
Markard, J., Wirth, S., Truffer, B., 2016. Institutional dynamics and technology legitimacy – a framework and a case study on biogas technology. Res. Policy 45, 330–344. https://doi.org/10.1016/j.respol.2015.10.009.
Martin, H., Coenen, L., 2015. Institutional context and cluster emergence: the biogas industry in Southern Sweden. Eur. Plan. Stud. 23, 2009–2027. https://doi.org/10.1080/09654313.2014.960181.
Mattes, J., 2012. Dimensions of proximity and knowledge bases: innovation between spatial and non-spatial factors. Reg. Stud. 46, 1085–1099. https://doi.org/10.1080/00343404.2011.552493.
Mattes, J., Huber, A., Koehrsen, J., 2015. Energy transitions in small-scale regions: what we can learn from a regional innovation systems perspective. Energy Policy 78, 255–264. https://doi.org/10.1016/j.enpol.2014.12.011.
McDowall, W., Ekins, P., Radošević, S., Zhang, L., 2013. The development of wind power in China, Europe and the USA: how have policies and innovation system activities co-evolved? Technol. Anal. Strateg. Manag. 25, 163–185. https://doi.org/10.1080/09537325.2012.759204.
Musiolik, J., Markard, J., 2011. Creating and shaping innovation systems: formal networks in the innovation system for stationary fuel cells in Germany. Energy Policy 39, 1909–1922.
Musiolik, J., Markard, J., Hekkert, M., Furrer, B., 2018. Creating innovation systems: how resource constellations affect the strategies of system builders. Technol. Forecast. Soc. Change. https://doi.org/10.1016/j.techfore.2018.02.002.
Panetti, E., Parmentola, A., Wallis, S.E., Ferretti, M., 2018. What drives technology transitions? An integration of different approaches within transition studies. Technol. Anal. Strateg. Manag. 108, 1–22. https://doi.org/10.1080/09537325.2018.1433295.
Planko, J., Cramer, J., Hekkert, M.P., Chappin, M.M.H., 2017. Combining the technological innovation systems framework with the entrepreneurs' perspective on innovation. Technol. Anal. Strateg. Manag. 29, 614–625. https://doi.org/10.1080/09537325.2016.1220515.
Quitzow, R., 2015. Dynamics of a policy-driven market: the co-evolution of technological innovation systems for solar photovoltaics in China and Germany. Environ. Innov. Soc. Transit. 17, 126–148. https://doi.org/10.1016/j.eist.2014.12.002.
Späth, P., Rohracher, H., 2012. Local demonstrations for global transitions: dynamics across governance levels fostering socio-technical regime change towards sustainability. Eur. Plan. Stud. 20, 461–479.
Stephan, A., Schmidt, T.S., Bening, C.R., Hoffmann, V.H., 2017. The sectoral configuration of technological innovation systems: patterns of knowledge development and diffusion in the lithium-ion battery technology in Japan. Res. Policy 46, 709–723. https://doi.org/10.1016/j.respol.2017.01.009.
The Wind Power, 2019. Germany Wind Farms Database. Tournefeuille Accessed 2019. https://www.thewindpower.net/.
trend:research, 2017. Eigentümerstruktur: Erneuerbare Energien: Entwicklung der Akteursvielfalt, Rolle der Energieversorger, Ausblick bis 2020. https://www.trendresearch.de/studie.php?s=672.
van Welie, M.J., Truffer, B., Yap, X.-S., 2019. Towards sustainable urban basic services in low-income countries: a technological innovation system analysis of sanitation value chains in Nairobi. Environ. Innov. Soc. Transit. https://doi.org/10.1016/j.eist.2019.06.002.
Weber, K.M., Truffer, B., 2017. Moving innovation systems research to the next level: towards an integrative agenda. Oxford Rev. Econ. Policy 33, 101–121. https://doi.org/10.1093/oxrep/grx002.
Wieczorek, A.J., Hekkert, M.P., Coenen, L., Harmsen, R., 2015. Broadening the national focus in technological innovation system analysis: the case of offshore wind. Environ. Innov. Soc. Transit. 14, 128–148. https://doi.org/10.1016/j.eist.2014.09.001.
Wilson, C., 2012. Up-scaling, formative phases, and learning in the historical diffusion of energy technologies. Energy Policy 50, 81–94. https://doi.org/10.1016/j.enpol.2012.04.077.
Wind, F.A., 2018. Ausbausituation der Windenergie an Land im Herbst 2018: Auswertung windenergiespezifischer Daten im Marktstammdatenregister für den Zeitraum Januar bis September 2018. Fachagentur Windenergie an Land, Berlin. https://www.fachagentur-windenergie.de/fileadmin/files/Veroeffentlichungen/FA_Wind_Zubauanalyse_Wind-an-Land_Herbst_2018.pdf.
Wüstemeyer, C., Madlener, R., Bunn, D.W., 2015. A stakeholder analysis of divergent supply-chain trends for the European onshore and offshore wind installations. Energy Policy 80, 36–44. https://doi.org/10.1016/j.enpol.2015.01.017.
Wüstenhagen, R., Wolsink, M., Bürer, M.J., 2007. Social acceptance of renewable energy innovation: an introduction to the concept. Energy Policy 35, 2683–2691. https://doi.org/10.1016/j.enpol.2006.12.001.
Yin, R.K., 2009. Case Study Research: Design and Methods, 4th ed. Sage, Thousand Oaks, Calif, pp. 219 pp.
Zhang, F., Gallagher, K.S., 2016. Innovation and technology transfer through global value chains: evidence from China's PV industry. Energy Policy 94, 191–203. https://doi.org/10.1016/j.enpol.2016.04.014.